

Twelve Reasons Why the Green Home Market Is Ready to Surge



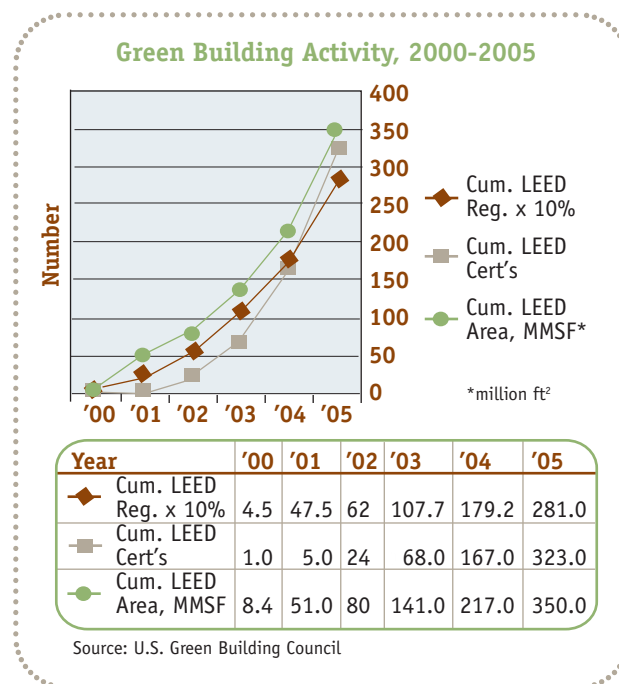
Jerry Yudelson, PG, MEA, chairs the U.S. Green Building Council's Greenbuild International conference and expo, the world's largest green building gathering. He is the author of *The Insider's Guide to Marketing Green Buildings* (available at www.yudelson.net).

If you need convincing about the marketability of green building, author Jerry Yudelson offers up a dozen reasons why green is the way to go.

1. The commercial and institutional green building market continues to grow at more than 50% per year (see Figure 1). In 2005, LEED-registered projects and project area grew by more than 50%, and LEED-certified projects grew by nearly 100%. LEED statistics indicate considerable growth potential for the green home market. Some of the LEED projects are multifamily residential structures, typically above three stories.

2. The new federal energy bill (Energy Policy and Conservation Act of 2005, or EPAct; see Table 1), which provides increased incentives for residential solar systems; prolonged oil prices above \$60 per barrel; and natural gas prices above \$8–\$10/MMBtu (\$0.80–\$1.00 per therm) have changed the psychology of the consumer for the first time since the oil price shocks of the 1970s. New credits for home builders should spur more investment in energy-efficient homes.

3. In November 2005, DOE's Energy Information Administration raised its projected 2025 oil prices, in today's dollars, from \$33 (2004 forecast) to \$54 per barrel, a 65% increase. Over time, this will probably translate into higher electricity and gas prices for residential applications, and more interest on the part of home buyers and homeowners in investing in



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conservation. In 2003, well before the current rise in energy prices, market studies for the King-Snohomish Master Builders Association (Seattle area) showed a willingness on the part of home buyers to pay 1% more—about \$2,500 on a new \$250,000 home—for a home energy package. Isn't it likely that home buyers will soon be willing to pay \$5,000 more, especially with the new \$2,000 home builders' incentive?

4. The continued movement of baby boomers back into urban cores will bring more discriminating buyers to condo developments, requiring builders to have a green point of differentiation. The companies that employ these boomers will want to offer greener office buildings to build a recruitment and retention edge. What people learn from working in green office buildings will also translate to their choices at home. The rise of the creative class, first chronicled in a book by the same name by Richard Florida in 2002, has the potential to change American demographic geographic patterns as

dramatically as the rise of Levittown and the suburban lifestyle did after World War II. Today, that pattern has begun to reverse itself. People want connectedness, they want the amenities of urban living, and they *don't* want to commute for hours each day for the privilege of mowing a patch of grass on Saturdays. This trend alone will lead to more energy-efficient homes and remodels, with a heavy focus on existing urban landscapes.

5. These same boomers will want to upgrade their single-family homes to make them energy efficient,

both to save on future utility costs and to show a concern for such issues as global warming and environmental protection. The strong role played by Governor Arnold Schwarzenegger's solar programs in promoting solar energy should help to kick-start the solar industry in California, for example. With the new solar PV incentives, look for a rapid rise in 1kW solar-electric systems and \$5,000 solar water-heating systems, as homeowners discover that installing these systems is the most visible way to show that they are doing something to save energy. (Right now, there is a shortage of semiconductor-grade silicon for producing PV panels on the market, but this shortage should be over by next year.) (See "Solar Subsidies are Booming," p. 6, for more on solar.)

6. A growing body of successful green home developments with a strong focus on solar and conservation features, in all major growth regions—including Florida, California, and the rest of the Sunbelt—will give developers confidence in their ability to deliver a high-

performance green development on a conventional budget. A good example is Shea Homes, in San Diego, California. Shea is the country's tenth largest builder, and in 2001 it developed a good package of energy conservation and solar technologies. Shea's new product line, the High-Performance Home, meets the

addition to these energy efficiency measures, High-Performance Homes incorporate passive-solar thermal water heating, and PV for electricity production. Shea Homes joined the industry's Built Green program in 2005—further evidence of its commitment to building energy-efficient housing.

roll out its finalized program guidelines in 2007. Given the success of the LEED for New Construction (LEED-NC) program and the growing recognition of the LEED brand name, LEED-H should begin to affect the residential market in significant numbers by 2008. Other local programs, such as the home-builders' Built Green program (which is now in seven states) and the programs of certain local utilities, as well as the National Association of Home Builders (NAHB) voluntary certification program, should also keep the new-home energy conservation market growing rapidly.

9. Related green buying habits should begin to affect the home-buying and retrofitting market. Look at the impact that the \$228 billion LOHAS market has had on organic foods, hybrid vehicles, ecotourism, and organic cotton, to name just a few examples. If home energy conservation could be marketed as more of a consumer product than a technical product, imagine how green homes could benefit from this growing consumer trend.

10. More cities will begin to require that standard commercial projects be built green—especially projects that have a major impact on the infrastructure. These requirements and policy directions will spill over into the home-building market over the next half decade. In 2005, many states and quite a few cities began to require LEED Silver level certification from their own buildings. One can view the growing tendency of home builders to undergo voluntary certification programs as an attempt to forestall legislative action on the part of states and cities, and one can predict that the green building trend will overwhelm that attempt over the next five years. The success of LEED-H will begin to make itself felt in the 2008–2010 time frame as a tool for green building advocates to use in persuading local governments to adopt green building requirements in all new homes.

11. Look for Energy Star to marry up with LEED to promote energy-efficient and zero-net-energy, or carbon-neutral buildings. We will begin to see

Table 1. EPA 2005: Key Provisions for Commercial Buildings

Affected Technology	Tax Credit
Photovoltaics	30% (residential limit is \$2,000 credit)
Solar thermal systems	30% (residential pool credits eliminated)
Microturbines	10% (up to \$200/kW credit)
Energy conservation investments for HVAC, envelope, lighting, and water-heating systems	\$1.80/ft ² (federal tax deduction if exceeding 50% savings vs. ASHRAE 90.1-2001 standard); up to \$0.60/ft ² for lighting retrofits alone
New homes exceeding 50% energy savings vs. model code	\$2,000 credit for site-built homes

Source: www.fsec.ucf.edu/EPAct-05.htm.

Table 2. LOHAS Consumer Markets, 2006

Sustainable Economy	Ecological Lifestyles	Alternative Health Care	Healthy Lifestyles	Personal Development
Green building and industrial goods	Ecological home and office products	Health and wellness solutions	Natural, organic food and beverage	Mind, body, and spirit products (CDs, books, tapes)
Renewable energy and alternative transportation	Organic/recycled fiber products	Acupuncture, homeopathy, naturopathy	Nutritional products	Personal development seminars
Resource-efficient products	Eco-friendly appliances	Holistic disease prevention	Dietary supplements	Yoga, fitness, weight loss
Socially responsible investing	Ecotourism and travel	Complementary medicine	Personal care	Spiritual products and services
U.S. Market: \$76.5 billion	U.S. market: \$81.2 billion	U.S. market: \$30.7 billion	U.S. market: \$30 billion	U.S. market: \$10.6 billion

Source: Natural Marketing Institute

requirements of an Energy Star Home, meaning that it is designed to use 30% less energy for heating, cooling, and water heating than would the same footprint built to 1993 National Model Energy Code standards. These homes are fitted with advanced features, including radiant-barrier roof sheathing that reflects heat away from the attic and thermostatic expansion valves that are designed to improve HVAC system performance. In

Atlanta, Chicago, Boston, New York, Seattle, San Francisco, and Portland, Oregon. This segment of the population is especially well represented in the Lifestyles of Health and Sustainability (LOHAS) psychographic market grouping, which is said to encompass up to 30% of the U.S. population (see Table 2). Of this market grouping, 65% are women.

8. The LEED for Homes (LEED-H) program, now in its pilot phase, expects to

buildings routinely cut energy use to 50% or more below current state standards through integrated design and innovative technological approaches. As we become more aware of the carbon problem, and of the ways in which buildings and urban settlement patterns contribute to global warming, architects and others in the design and construction industry will begin to face up to their responsibility to find ways to address these issues. One sign of this change is the position statement adopted by the American Institute of Architects (AIA) in December 2005, calling for a minimum 50% reduction in building energy consumption by 2010. In its statement, the AIA supported “the development and use of rating systems and standards that promote the design and construction” of more resource-efficient communities. As public companies, the major home

builders will have to become more socially responsible, if they want to get their projects permitted, built, and sold—and if they want to attract top talent to keep their revenues and profits growing. Look for the corporate governance and socially responsible investing movements to begin to influence how the top ten builders (who account for 30% of all new homes in the country) plan, design, and market their homes.

12. The U.S. Green Building Council’s LEED green building rating system registered more than 1,000 projects last year for the first time ever, totaling more than 130 million ft² of space. I predict that the total number of LEED-registered projects will increase more than fourfold by the end of the decade, and that it will continue to increase at more than 20% per year thereafter. I further predict that the number of LEED-certified projects will exceed 500 by the end of 2006. This means that homeowners and home buyers everywhere will continue to see more information about

green buildings in their cities and towns. I believe that this will translate into significantly increased activity in the home energy markets, both for new homes and for conservation retrofits.



FOR MORE INFORMATION:

For figures on projected 2025 oil prices, see U.S. Department of Energy, Energy Information Administration, Annual Energy Outlook 2006 at www.eia.doe.gov/oiaf/aeo/aeoref_tab.html.

For a PATH evaluation report, see the Toolbase Web site at www.toolbase.org/tertiaryT.asp?DocumentID=4120&CATEGORYID=1505.

For more information on green marketing, see the Natural Marketing Institute Web site at www.nmisolutions.com.

To view a copy of the AIA’s position statement, go to www.aia.org.